PwC's 2025 Digital Trends in Operations Survey

Accelerating the shift from triage to transformation

91%

of operations and supply chain leaders say they'll significantly change supply chain strategies because of US trade policy changes

57%

have integrated AI into selected functions or throughout their organization

92%

say tech investments haven't fully delivered the expected results

With geopolitical volatility, surging costs and other disruptions, the question isn't whether operations and supply chains will change. It's how fast and intentionally leaders can guide that evolution.

PwC's 2025 Digital Trends in Operations Survey — based on insights from 610 operations and supply chain leaders — paints a clear picture. COOs and their teams play a critical role in driving business transformation, but many are still juggling short-term firefighting with long-term strategy.

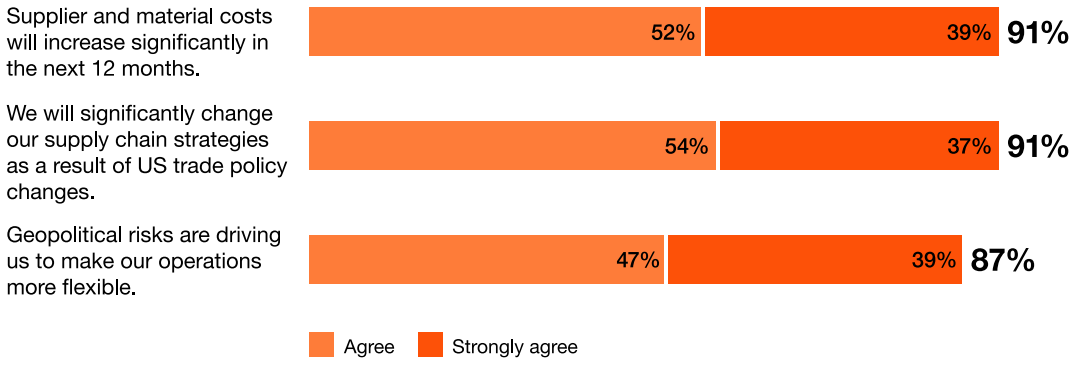
Here's what we found and what you can do to move faster from a triage mindset to accelerating transformation in the months and years ahead.

Finding the right balance is essential but elusive

The tension between rapid response and long-range planning is real. A large majority — 82% — say they face challenges in balancing short-term needs with long-term strategic changes. Most (68%) say the challenges are manageable, and 14% say they're significant. Still, executing on a multiyear strategy while prioritizing ongoing cost and efficiency pressures — especially with sweeping tariff changes under the Trump administration — uses time and resources that could be dedicated elsewhere.

And current events are causing concerns for many operations and supply chain leaders. Roughly nine out of 10 say supplier and material costs will increase significantly in the year ahead. That can strain budgets, reduce profit margins, trigger supplier renegotiations and disrupt planning. A similar number of respondents (91%) say US trade policy changes are moving them to significantly change supply chain strategies, and almost as many (87%) say geopolitical risks are driving them to more flexible operations. All this can raise the pressure for more effective scenario planning and adaptability in supply chains.

Near-term issues expected to impact operations and supply chain – now and going forward



Q: To what extent do you agree or disagree with the following statements? (Response to 'Agree' and 'Strongly Agree'.)
Base: 610
Source: 2025 Digital Trends in Operations Survey

Artificial intelligence (AI) could be part of the solution. AI can accelerate the speed of insights, decision-making and organizational change. We're also beginning to see how **AI agents** can fundamentally transform work and change how businesses operate. For instance, AI agents **can be coordinated** across demand forecasting, procurement and logistics tracking to address potential supply chain bottlenecks and reduce delays.

More than half (53%) of survey respondents say AI is being used in either a few areas or widely to anticipate and mitigate supply chain disruptions. Another 31% say they're testing and piloting AI for those purposes. Responses are similar for scenario planning and operational transparency, with 55% using it in at least a few areas and 29% testing and piloting AI.

What it means for you:

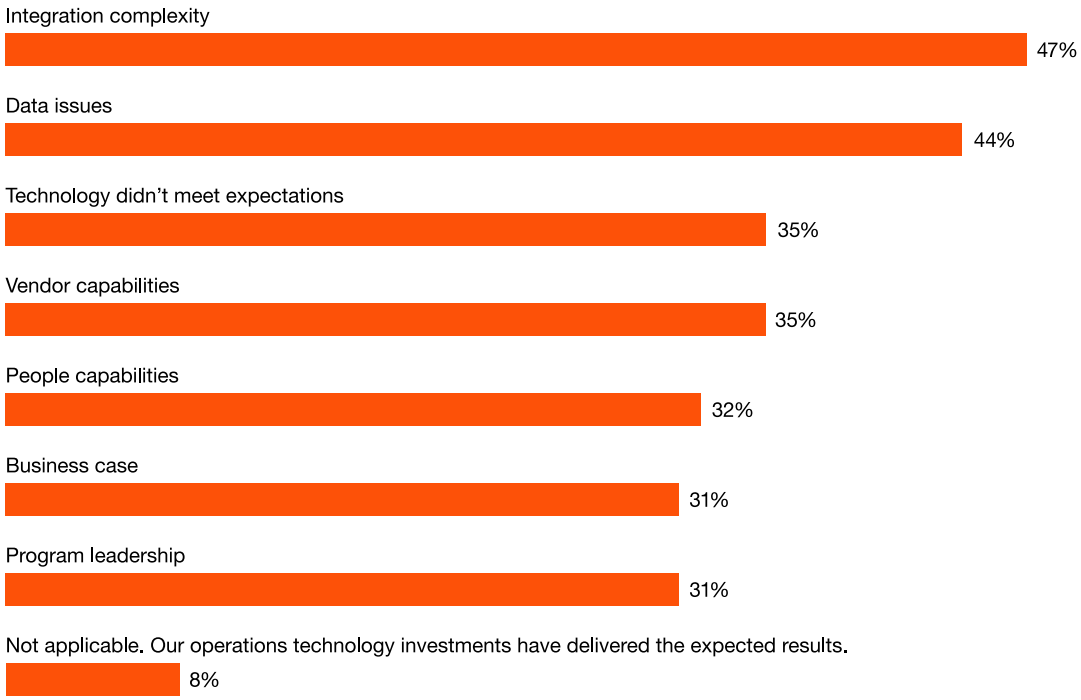
Focus on being efficient in the near term but do it in a way that allows your company to be flexible and ready for future challenges. Explore how to apply AI in different ways — from driving efficiencies today to anticipating and evaluating scenarios to better configure for tomorrow.

02

Cracking the complexity conundrum in tech adoption

Among technologies used by companies, AI (59%) and cloud (56%) are tops, and almost all of those respondents say those capabilities are somewhat or very effective in creating value (98% for AI, 96% for cloud) — driving revenue, increasing productivity or managing costs. Advanced capabilities like digital twins and data ecosystems lag far behind in adoption but could be a missed opportunity. Although only 21% say their companies use digital twins, 97% of those respondents say that capability is either somewhat or very effective in creating value.

Here's why tech investments haven't delivered the expected results



Q: If your investments in operations technology have not fully delivered the expected results, which of the following are the reasons why? (Select all that apply.)
Base: 610
Source: 2025 Digital Trends in Operations Survey

There's other cause for concern. Despite using multiple digital capabilities, 92% of operations and supply chain leaders cite at least one reason why tech investments haven't fully delivered the expected results, and 83% cite two or more reasons. Integration complexity (selected by 47%) and data issues (44%) are the most common reasons — a shift from the previous survey and likely an indicator of the struggles of adapting to and leveraging new AI solutions. Alarmingly, one of the least selected reasons is the business case for tech investment — unclear objectives, weak rationale or difficulty understanding costs — which may suggest a blind spot for many companies.

What it means for you:

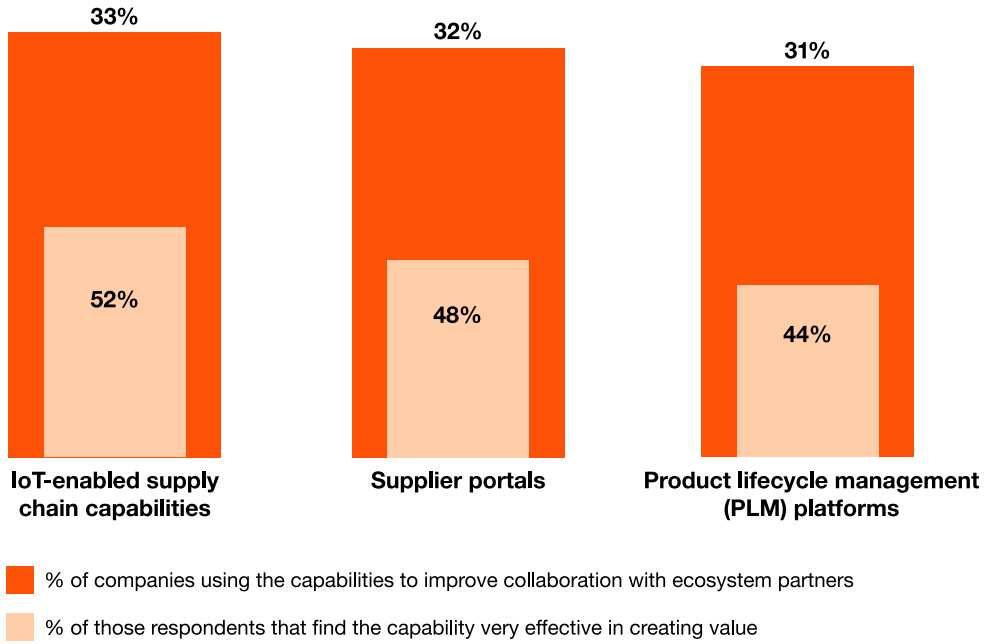
As technology becomes more complex and issues more frequent, it's increasingly important to understand how a capability integrates with others. Value comes from using tech well, which often means investing in new architectures. Less used capabilities like digital twins show high impact and can answer many "what if" questions when applied effectively — especially in an environment where organizations are looking to become more flexible in their operations. Focus less on quantity and more on integration and measurement, especially in tracking costs and ROI.

03

Reconfigure supply chains to be more adaptable and insightful

The turbulence so far in 2025 makes it clear: Your supply chain needs to be faster, smarter and more resilient than ever. As COOs and supply chain leaders increasingly evaluate their entire ecosystems, they're using several technologies to improve collaboration with vendors, customers, third-party logistic providers and other value chain partners.

Some less used capabilities are effective in creating value



Q: Which digital capabilities does your company use to improve collaboration with ecosystem partners (e.g., vendors, customers, intermediaries, 3rd party logistic providers)? (Select all that apply. "None" not shown.)
Q: And, how effective are these capabilities in creating value? (e.g., driving revenue, increasing productivity, managing costs) (Response to 'Very effective'.)
Base: 610
Source: 2025 Digital Trends in Operations Survey

Some lesser used capabilities are delivering solid returns and could help create value for companies. Consider that only 33% are using Internet of Things (IoT) enabled supply chain capabilities, yet 52% of those respondents say they've been very effective in creating value.

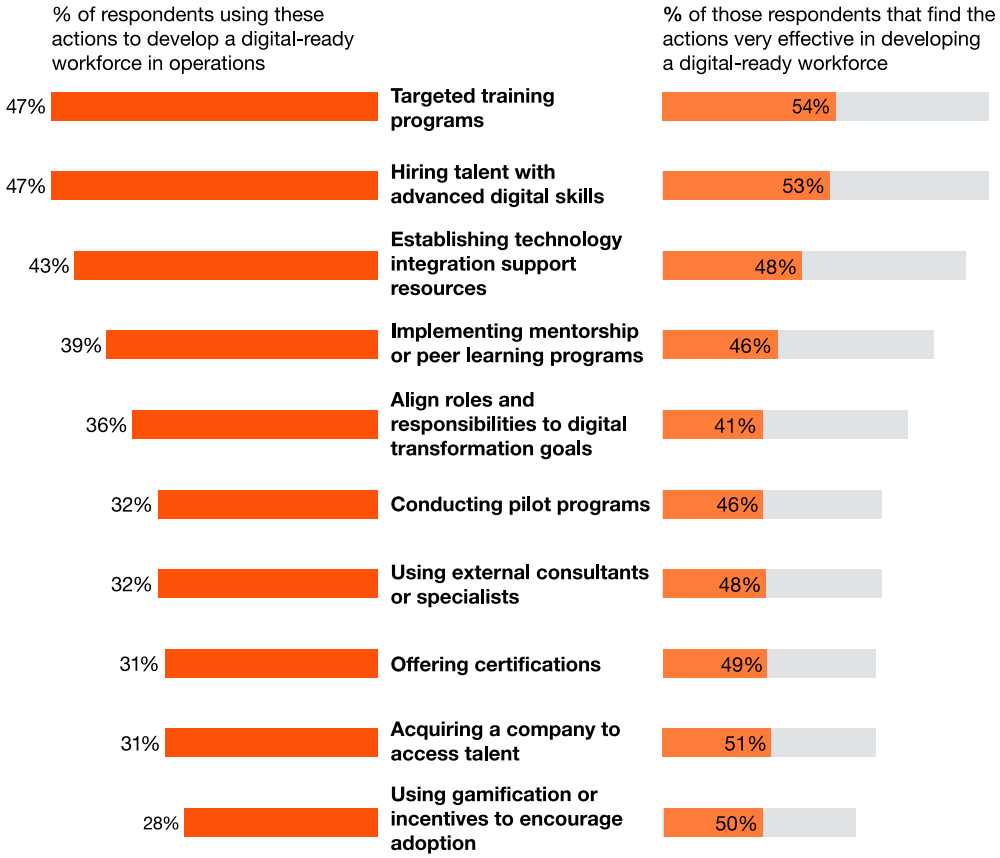
What it means for you:

Maintain focus and a results-oriented mindset amid the growing volume and complexity of new technologies. Keeping an eye on "fit for purpose" is key, and solutions that can improve visibility, transparency and scenario modeling should be investigated considering how effective they are for other companies.

Go beyond training and hiring to build a digital-ready workforce

Companies are trying multiple ways to develop a digital-ready workforce in operations, and hiring skilled talent and targeted training (47% each) top the list. But similar to improving collaboration with ecosystem partners, some less-used actions are proving very effective and could be a differentiating opportunity for companies.

Training and hiring are leading moves to develop a digital-ready workforce, but others also are very effective



Q: How is your company developing a digital-ready workforce in operations? (Select all that apply.)
Q: And, how effective are all these actions in developing a digital-ready workforce in operations? (Response to 'Very effective'.)
Base: 610
Source: 2025 Digital Trends in Operations Survey

Fewer than one-third of the respondents say they're using gamification or incentives, offering certifications or acquiring a company for talent to build a digital-ready workforce. Yet among the companies doing those things, about half of survey respondents say they've been very effective.

What it means for you:

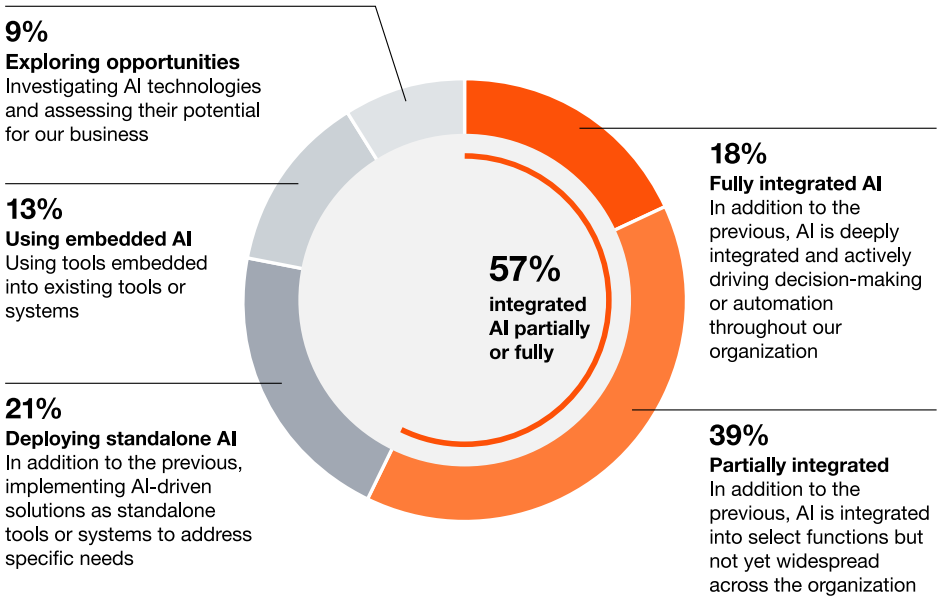
Hiring and training will always be important, but those are table stakes. To gain a competitive edge, consider specific incentives and certifications to boost your digital workforce. Also don't sleep on potential acquisitions that could provide an influx of tech expertise.

AI as a cornerstone of enabling your digital strategy

AI isn't just another technology, and this is an adjustment from how executives have historically thought about AI — as solely an enabler that's primarily the domain of IT people. As it continues to become a natural part of everything people do, AI should be intrinsic to your strategy and capability systems — including operations.

The survey found that many companies are using AI in operations and supply chains, with 57% of respondents saying they've already integrated AI partially or fully into their operations.

AI integration at companies



*Note: Totals may not add up due to rounding. Response to 'Unsure' not shown.
Q: Which of the following best describes how your company is approaching AI integration? (Select one.)
Base: 610
Source: 2025 Digital Trends in Operations Survey

But adoption often meets with friction, and several challenges remain in effectively scaling AI in operations. Integration with existing systems — ranked among the top three challenges by 42% of respondents — and data issues (37%), such as availability and quality, are most common. This aligns with our experience as we often see companies overlook the complexity in developing new AI capabilities — all of which require solid integration and data to power high ROI solutions. When integrated well, AI can be a tremendous help in resolving data challenges.

What it means for you:

AI is everywhere, and scaling it in operations can be a challenge. If you're already on the road of integration, explore how AI can help address data challenges, focus on evolving how employees work and enable them to help drive more change. If your AI program is still young, be open to learning from other organizations and build a culture that allows your teams to absorb and advance your AI aspirations.

Survey insights by industry

Consumer markets

Consumer markets (CM) companies are moving from AI experimentation to excellence. Our survey found that both AI and machine learning and **AI agents** are among the top three digital capabilities currently deployed at CM companies.

Crucially to CM operations in this volatile environment, 48% of respondents also note using AI agents to improve collaboration with their ecosystem partners, and 59% of respondents say AI agents are very effective at creating value, which is the highest rate of any industry in the survey. We see using AI agents in their ecosystem collaborations as a necessity going forward, so it's not surprising to see that this capability already is a focus.

CM companies note three ways they're **keeping employees up to date on fast moving AI innovations**: Hiring talent with advanced skills (50%), implementing targeted training programs (46%) and establishing technology integration support resources (43%).

There have been setbacks. CM companies cite the biggest challenges to scaling their AI usage in their operations were integration with existing systems (40% say it's in the top three), data issues (39%) and difficulty predicting and planning for cloud costs (35%).

Our biggest takeaway is that this is just the start. Only 14% of CM respondents say AI is deeply integrated and actively driving decision-making or automation throughout their company. We expect **AI will be a driving force in CM in the coming years**, and the winners in the market will see AI become intrinsic to almost everything they do, including operations.

Energy, utilities and resources

As digital transformation continues to reshape global operations, the energy, utilities and resources (EUR) sector finds itself navigating a complex mix of challenges and opportunities. Companies are making steady strides toward digital maturity but still face hurdles in aligning short-term needs with long-term strategic goals.

Among the most pressing priorities for energy companies are increasing operational efficiency (93%), developing a digital-ready workforce (92%) and achieving sustainability goals (90%) — with all percentages reflecting those who cite each as a moderate or high priority.

But EUR companies differ from other industries in considering some issues and goals as a high priority. While increasing efficiency is widely acknowledged as critical, for instance, only 35% of EUR respondents say it's a high priority, compared to 48% across all sectors. Meanwhile, 58% of EUR leaders cite it as a moderate priority, compared to 43% overall. That may suggest strong intent but a cautious approach among EUR companies when it comes to resource allocation.

Industrial products

Industrial products (IP) companies are embracing digital transformation, even as they navigate a complex balancing act between short-term pressures and long-term goals. Most are making strategic digital bets. Our survey indicates that all IP companies are using digital capabilities, and among those that are using AI and machine learning and cloud-based technology, a slight majority — 54% for AI and machine learning, and 53% for cloud — say they're very effective in creating value.

Yet that progress is tempered by real challenges. Four out of five IP leaders say they face difficulties in balancing today's needs with long-term goals. Cost control ranks especially high, with 91% of IP respondents prioritizing it and 54% citing it as a high priority.

This emphasis comes as 95% of IP respondents expect supplier and material costs to increase significantly, prompting plans for supply chain shifts and resiliency measures. Notably, 35% say cost management is primarily a long-term goal, signaling a shift beyond near-term firefighting toward more sustainable transformation.

Pharma and life sciences

Pharmaceutical, life sciences and medical technology companies are watching the evolving policy landscape and analyzing its anticipated impact on the supply chain. Most leaders (89%) agree or strongly agree they will significantly change their supply chain strategies as a result of US trade policies, and almost as many (87%) say

the same for supplier and material costs increasing significantly in the next 12 months.

No strangers to disruption, industry leaders are preparing for the challenges ahead. Almost all (98%) say that digital tools have improved visibility into their end-to-end operations. More than half (51%) are using predictive analytics — compared with 38% of all respondents — to improve collaboration with ecosystem partners like vendors, customers, intermediaries and third-party providers.

Nearly three quarters (73%) say that while they face challenges in balancing short-term needs with longer-term strategic changes, these challenges are manageable. They're leaning into AI to improve operations, and 62% believe these tools are very

Tech and telecom

Nearly all tech and telecom operations and supply chain leaders (96%) say digital tools have improved visibility into their end-to-end supply chain costs. Operations leaders are leaning on AI, cloud and automation to streamline everything from quality control to distribution, squeezing out efficiency gains and managing costs.

Yet this digital drive isn't just about quick wins. Leaders are keenly aware of the long game — balancing immediate needs with future resilience and sustainability. Nearly three quarters (74%) admit it's a challenge to juggle short-term pressures with long-term goals, but they also say those challenges are manageable.

Tech and telecom companies are invested in AI integration across the enterprise, and most are using AI in some fashion. Still, many haven't reaped the full rewards of these investments due to integration complexities and issues with vendors — and with the quality, availability and security of data. The tech and telecom companies that are pulling ahead are marrying bold tech adoption with strategic vision — leveraging AI and newfound cost visibility to hit today's targets while building tomorrow's competitive edge.

Are you ready to take decisive action?

Reinventing your operations can provide a competitive advantage, and it's not too late to be proactive with your digital strategy for operations and supply chain. To outpace the pack, your company should make smart trade-offs, adopt capabilities that truly deliver and build a workforce and operating model equipped to thrive in uncertainty. Here are eight moves you can take now.

- **Anticipate versus react to disruption.** Using tech for basic risk management won't cut it. Leverage predictive models for supply chain disruptions and build flexible operating models that adapt in real time. Building your scenario planning capabilities is critical.

- **Invest in data as a strategic asset.** Data integration should be a core capability, not an IT problem. Remove departmental barriers, improve data quality and enable cross-functional access so new capabilities can generate more value.

- **Double down on making digital ROI clear**. Prioritize initiatives that improve cost visibility and reduce risk in operations. Tie AI and other tech investments directly to performance metrics and value drivers — and develop specific capabilities to model and manage costs, especially when applying AI.

- **Integrate, don't just implement.** Simplify system architecture and build a foundation for integration instead of treating it as an afterthought. By doing that, you can better unlock value from data and advanced decision models.

- **Make AI everyone's business.** Pilot programs have their purpose but can be limited in scope and impact. Focus on AI use cases like inventory optimization, connected worker and digital coach enablement, close loop analytics for operations excellence and ecosystem collaboration to deliver measurable value.

Find out more about our Operations Transformation services

Harness new technology, reimagine your business and accelerate outcomes.

Learn more

Explore full 2025 survey data

Access an interactive dashboard, filter by industry and more.

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- **Strengthen ecosystem/partner collaboration.** An effective ecosystem can help simplify complexity and better deliver for your customers. Build an ecosystem strategy that includes tech, data and value chain partners. Each can bring capabilities required to enable differentiated service and value for your customers. AI agents also can help reduce friction across your ecosystem.

- **Embed learning into your operating model.** Incorporate mentorship, role evolution and peer learning programs into your culture. Increasing digital fluency in operations and supply chains is critical to standing apart from competitors.

About the survey

PwC's 2025 Digital Trends in Operations Survey surveyed 610 operations executives and supply chain officers in February and March 2025. Respondents in the online survey included C-suite executives, upper management, directors and managers based in the US who either have sole responsibility for business decisions on operations and supply chain or procurement operations or share influence with others regarding those decisions. Sectors surveyed include consumer markets; energy, utilities and resources; pharmaceuticals and life sciences; industrial products; and technology and telecommunications.

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